Harmony Certificate Layer

Mathematical Reference, Implementation Map, and Expansion Roadmap

Companion notes for the Harmony toolbox
Track B = Lekić–Beerten impedance method ($H = YZ_{\mathrm{eq}}$)
`src/Solver/Certificate/` + `StabilityEstimate` + MMC

August 4, 2026

Abstract

These notes document what Harmony's certificate layer computes today, how it sits on the **Lekić–Beerten** impedance pipeline (Y, Z_{eq}, $H = YZ_{\mathrm{eq}}$), and what remains relative to full compositional DW/SRG interconnection theorems. Local **geometric certificates** (numerical range, small-gain, DW-lite shell, sector/loop-shift) are implemented in `Geometric_certificates.*`. Formulas, workflows, distinctive test cases, and how to run the unit/demo tests are included.

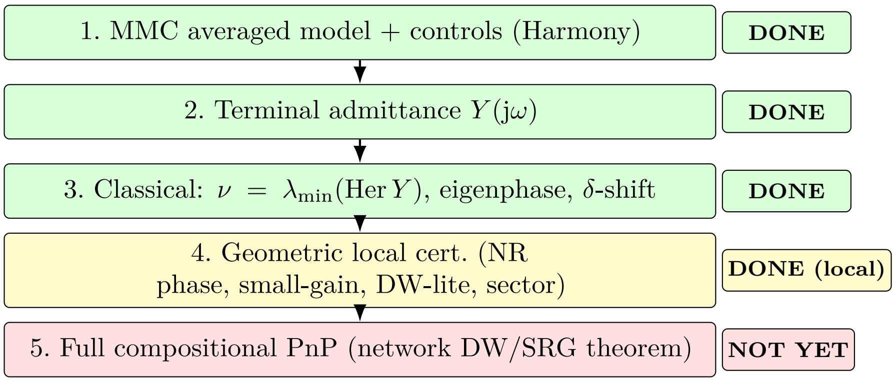
Contents

1 Ownership of the two tracks

Track B — Impedance / return ratio	**Track A — Local geometric certificates**
$Y(\mathrm{j}\omega)$, Z_{eq}, $H = YZ_{\mathrm{eq}}$, MIMO Bode/Nyquist. **Lekić–Beerten line** (author home turf). First-class in Harmony (`StabilityEstimate`).	Passivity, eigenphase, numerical-range (small-phase), small-gain, excess/shortage, DW-lite shell, sector/loop-shift, PnP gate. **Local sufficient screens**, not a full compositional DW/SRG network theorem.

Narrative: extend the H-based MMC–HVDC platform with modern local geometric certificates, and measure their conservatism against your own H — not "reimplement ETH theory."

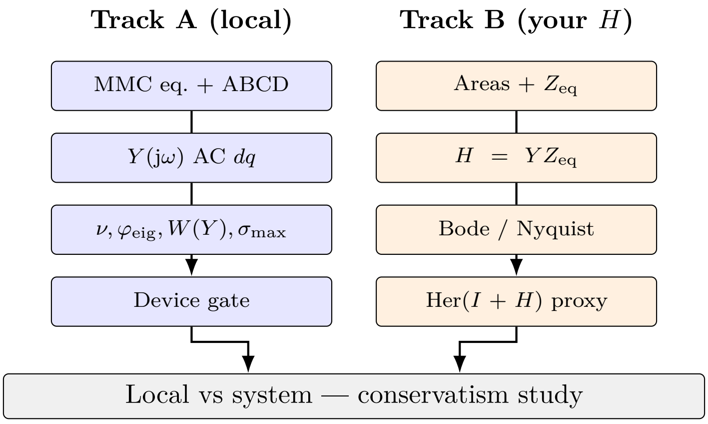
2 Pipeline vs Harmony today



Network assessment today uses Track B in parallel: $H = YZ_{\mathrm{eq}}$ (Bode/Nyquist) plus optional $\mathrm{Her}(I + H)$ proxy.

Step	In Harmony	Status
MMC model	Averaged MMC + ABCD + GFM	Done
$Y(s)$	`compute_y_parameters`	Done
Classical screens	ν, eigenphase, δ-shift	Done
Geometric local	NR, small-gain, DW-lite, sector	Done (local)
Network check	$H = YZ_{\mathrm{eq}}$ (+ $\mathrm{Her}(I + H)$)	Done (impedance)
Compositional PnP	Network DW/SRG theorem	Missing

3 Two parallel tracks



Code map. `Geometric_certificates`, `Stability_certificate`, `Device_gate`, `PnP_library`, `Operating_region`, `Tuning_assistant`, `Local_vs_system`, `StabilityEstimate`. Demo: `Harmony-cpp certificate_figures`.

4 Plant, linearization, admittance

4.1 Equilibrium and ABCD

$$\dot{x} = f(x,u), \quad y = g(x,u), \qquad f(x^\star, u^\star) = 0,$$
$$\Delta\dot{x} = A\Delta x + B\Delta u, \qquad \Delta y = C\Delta x + D\Delta u.$$

4.2 Admittance

$$Y(s) = C(sI - A)^{-1}B + D, \qquad s = \mathrm{j}\omega.$$

4.3 MMC multiport

$$Y = \begin{bmatrix} Y_{\mathrm{dd}} & Y_{\mathrm{da}} \\ Y_{\mathrm{ad}} & Y_{\mathrm{aa}} \end{bmatrix}, \qquad Y_{\mathrm{ac}} := Y_{\mathrm{aa}} \in \mathbb{C}^{2\times 2}$$

(default: `ac_dq_block=true`).

Remark 1 (Methodological home)**.** Forming $Y(\mathrm{j}\omega)$ and assessing interconnection via $H = YZ_{\mathrm{eq}}$ is the Lekić–Beerten impedance workflow (Track B).

5 Classical metrics

$$\mathrm{Her}(Y) = \tfrac{1}{2}(Y + Y^*), \qquad \nu(Y) = \lambda_{\min}(\mathrm{Her}\, Y), \qquad \nu_\delta(Y) = \lambda_{\min}(\mathrm{Her}(Y + \delta I)).$$
$$\varphi_{\mathrm{eig}}(Y) = \max_{i:\,\lambda_i \neq 0} |\arg \lambda_i(Y)| \quad \text{(heuristic eigenphase).}$$

Excess / shortage:

$$\nu_+(Y) = \max(0, \nu(Y)), \qquad \nu_-(Y) = \max(0, -\nu(Y)).$$

Log-grid f_k; worst-case aggregates over the sweep. ALLOW iff all enabled checks in `CertificateSpec` pass (`sweepPassesSpec`).

6 Geometric certificates (implemented)

Module: `Geometric_certificates.h/.cpp`. Metrics are always written into each `CertificateSweep`; enforcement is opt-in via `require_*` flags.

6.1 Numerical range (field of values)

$$W(Y) = \{v^*Yv : \|v\| = 1\} \subset \mathbb{C}.$$

Support-function boundary

$$z(\theta) = e^{\mathrm{j}\theta}\,\lambda_{\max}(\mathrm{Her}(e^{-\mathrm{j}\theta}Y)), \qquad \theta \in [0, 2\pi).$$

- $\min \operatorname{Re} W(Y) = \nu(Y)$.
- **NR phase:** $\varphi_{\mathrm{NR}}(Y) = \max_{z\in\partial W(Y)} |\arg z|$ (set to 180° if $0 \in W(Y)$).
- **Zero margin:** $m_0(Y) = -\min_{\theta} \lambda_{\max}(\mathrm{Her}(e^{-\mathrm{j}\theta}Y))$. Then $m_0 > 0 \Rightarrow 0 \notin W$; $m_0 \le 0 \Rightarrow 0 \in W$ (dense θ-grid).

Flags: `require_nr_phase`, `require_nr_zero_outside`.

6.2 Small-gain

$$\sigma_{\max}(Y) = \|Y\|_2, \qquad \text{pass if } \sigma_{\max}(Y) \le \gamma.$$

Flags: `require_small_gain`, `gain_limit`$=\gamma$.

6.3 Davis–Wielandt shell (DW-lite, local)

$$\mathrm{DW}(Y) = \big\{(v^*Yv, \|Yv\|^2) : \|v\| = 1\big\} \subset \mathbb{C} \times \mathbb{R}_+.$$

Harmony samples $\mathrm{DW}(Y)$ and plots the xz projection $(\operatorname{Re}(v^*Yv), \|Yv\|^2)$. Local DW-lite gate (`require_dw`):

$$0 \notin W(Y) \quad \text{and} \quad \varphi_{\mathrm{NR}}(Y) \le \varphi_{\lim}.$$

Remark 2 (Honesty)**.** Local geometric screen only — not the ETH projected-shell SDP / converter–network DW separation theorem.

6.4 Sector / loop-shift

For sector $[\alpha, \beta]$ with $\beta > \alpha$,

$$T = (Y - \alpha I)(\beta I - Y)^{-1}$$

(when invertible). Sector certificate: $\nu(T) \ge \varepsilon$ (`require_sector`, `sector_alpha`, `sector_beta`).

6.5 Spec flags

Flag	Enforces
`require_passivity`	$\nu_{\mathrm{worst}} \ge \varepsilon$
`require_phase`	$\varphi_{\mathrm{eig,worst}} \le \varphi_{\lim}$
`require_shifted`	$\nu_{\delta,\mathrm{worst}} \ge \varepsilon$
`require_nr_phase`	$\varphi_{\mathrm{NR,worst}} \le \varphi_{\lim}$
`require_nr_zero_outside`	$m_{0,\mathrm{worst}} > 0$
`require_small_gain`	$\sigma_{\max,\mathrm{worst}} \le \gamma$
`require_dw`	DW-lite (zero-out + NR phase)
`require_sector`	loop-shifted sector passivity

7 Track B: return ratio

$$H(\mathrm{j}\omega) = Y_{\mathrm{conv}}(\mathrm{j}\omega)\, Z_{\mathrm{eq}}(\mathrm{j}\omega).$$

Optional proxy:

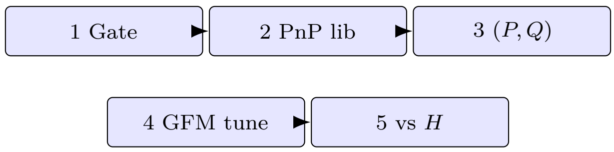
$$\nu_{\mathrm{sys}}(\omega) = \lambda_{\min}(\mathrm{Her}(I + H(\mathrm{j}\omega))).$$

Classical assessment remains Bode/Nyquist of H. Consistency:

$$\text{consistent} \iff (\text{local\_pass} = \text{system\_pass}).$$

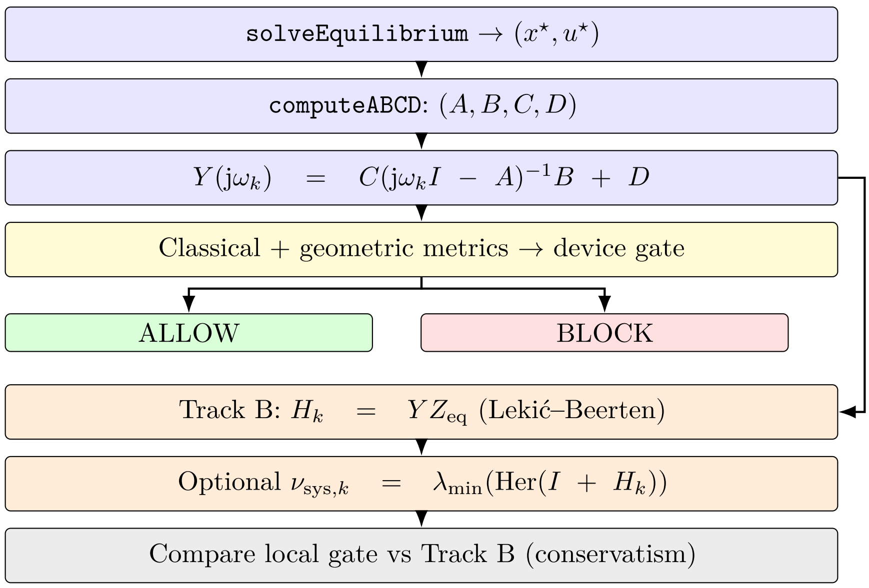
GFM-MMC often: local FAIL, system PASS — seed for non-passivity bands.

8 Five workflows



Gate → catalog; (P, Q) / OPF region; droop search; local-vs-H.

9 End-to-end flowchart



10 How to run tests

10.1 Unit tests (GoogleTest)

Source: `tests/testCertificate.cpp`. Executable: `testharmony` (from the `tests/` CMake project).

Suite	Covers (GTest filter)
Metrics	Classical ν, shifted passivity (`CertificateMetrics.*`).
Workflows	Device gate + PnP JSON; sweep sizes (`CertificateWorkflows.*`).
Geometric	NR / DW-lite / small-gain / sector (`GeometricCertificates.*`).

```
testharmony.exe --gtest_filter=GeometricCertificates*
testharmony.exe --gtest_filter=Certificate*:GeometricCertificates*
```

10.2 Integration / paper demo

```
Harmony --cpp certificate_figures --no-plot
Harmony --cpp certificate_figures
```

Expected on default GFM-MMC: local **BLOCK** (passivity FAIL); system $\mathrm{Her}(I + H)$ **PASS**; verdict **DISAGREE**.

Smoke test: `testexamples.cpp` calls `example_certificate_figures(false)`.

11 Distinctive test cases vs classical Beerten studies

Classical Beerten / early MMC impedance papers emphasized averaged MMC models, GFL-oriented controls, and impedance / eigenvalue / EMT validation of H — not decentralized geometric gates, GFM droop certificates, post-OPF (P, Q) regions, or open multi-case toolbox packs.

11.1 Cases Harmony / CRESYM can uniquely offer

Case family	What is new
GFM-MMC + local vs H	Outer-loop GFM; passivity/NR FAIL vs H PASS table. Demo: `certificate_figures`.
Non-passivity band maps	Contours of $\nu(f)$, $\nu_-(f)$, $\varphi_{\mathrm{NR}}(f)$, $\nu(P, Q)$, $\nu(K_P, K_Q)$; overlay H-margins.
Post-OPF certified (P, Q)	OPF setpoint + certificate region around $(P^\star, Q^\star)$.
MTDC / multi-MMC PnP	ALLOW/BLOCK catalog; recheck gates vs H at each cut.
AC/DC multiport cuts	Certify Y_{aa}, Y_{dd}, full Y; AC and DC H.
Weak AC / topology	SCR / topology sweeps: when local geometry disagrees with H.
Open reproducible pack	JSON/C++ cases + CSV sweeps (software-paper asset).
EMT / RTDS twin	Contested OPs where PnP blocks but H passes.

11.2 Minimal paper case pack

1. **C0** Stiff-grid GFM-MMC (`mmc_gfm` / `certificate_figures`) — baseline disagreement.
2. **C1** Same MMC, weak SCR sweep (e.g. SCR = 2–10).
3. **C2** Point-to-point HVDC with AC and DC H cuts.
4. **C3** Three-terminal MTDC: add/remove one converter.
5. **C4** OPF-feasible (P, Q) cloud + certificate overlay.
6. **C5** (optional) EMT check of 2–3 contested points.

12 Literature map

Passivity / phase: excess passivity, small-phase, numerical range; He–Dörfler dVOC passivity (TPWRS).

GFM parametric PnP / loop-shifting: Häberle–He–Huang–Dörfler–Low, arXiv:2503.05403; dynamic multipliers arXiv:2602.09150.

DW-shell / SRG (full interconnection): arXiv:2508.17033; arXiv:2507.19918; Baron-Prada–Anta–Dörfler SRG arXiv:2601.16014.

IQC / multipliers: IQC, Zames–Falb, dissipativity.

Track B: Lekić–Beerten / Sakinci–Lekić–Beerten $H = YZ_{\text{eq}}$ — gold-standard comparator.

Expansion order: (1) NR/DW-lite conservatism on C0–C1 vs H; (2) weaker local rules on non-passivity bands; (3) full network DW/SRG only if compositional PnP claims are required; (4) always validate vs H.

13 GFM non-passivity bands (theory target)

$$\Omega_{\text{bad}} = \{\omega : \nu(Y_{\text{ac}}(\mathrm{j}\omega)) < \varepsilon\}.$$

Seek property C on Ω_{bad} such that $C(Y)$ + network class $\mathcal{Z}$ implies interconnection OK (consistent with H/Nyquist). Geometric metrics already expose ν_-, φ_{NR}, and m_0 over frequency for this programme.

14 Regenerate evidence

```
Harmony --cpp certificate_figures
Harmony --cpp certificate_figures --no-plot
Harmony --cpp mmc_gfm
testharmony.exe --gtest_filter=GeometricCertificates*
```

Outputs under `files/`: sweeps (incl. NR phase, zero-margin, small-gain, shortage), operating-region CSV, PnP library JSON.

15 Symbol table

Symbol	Meaning
Y	Device terminal admittance
Y_{ac}	AC dq block
ν	$\lambda_{\min}(\operatorname{Her} Y)$
ν_+, ν_-	excess / shortage passivity
φ_{eig}	max eigenphase (heuristic)
φ_{NR}	numerical-range phase (small-phase)
$W(Y)$	numerical range / field of values
m_0	NR zero-margin ($> 0 \Rightarrow 0 \notin W$)
$\sigma_{\max}$	operator norm (small-gain)
$\mathrm{DW}(Y)$	Davis–Wielandt shell
Z_{eq}	Network equivalent at cut
H	YZ_{eq} (Lekić–Beerten)
ν_{sys}	$\lambda_{\min}(\operatorname{Her}(I + H))$ proxy